

CAAM/STAT31470: Homework #2

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: In this week's problems, we look at the properties of analytic functions, and how to determine when (and where) functions are analytic. We will also explore some intuition around what makes some functions not analytic, and what happens at the boundary of analyticity.

Homework is due: **Wednesday April 9, 11:59pm.**

Problem 2.1 (6 points)

Determine whether the following functions are analytic, and specify on what domains.

- (a) $f(z) = 3x + y + i(3y - x)$
 - (b) $f(z) = 2xy + i(x^2 - y^2)$
 - (c) $f(z) = \frac{z^2 + 1}{(z+2)(z^2 + 2z + 2)}$
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Problem 2.2 (6 points)

We have seen that functions containing a conjugate of z , are typically not analytic. In fact, we can show that this is the case (at least formally) using the following argument.

Let $f(z) = u(x, y) + iv(x, y)$ and suppose $f(z)$ is analytic.

- (a) For the moment, suppose that z and $\bar{z}$ are independent variables. Then using the change of variables

$$x = \frac{1}{2}(z + \bar{z}), \quad y = \frac{1}{2i}(z - \bar{z})$$

to find the partial derivatives $\partial u/\partial z$ and $\partial u/\partial \bar{z}$ in terms of the partial derivatives of $\partial u/\partial x$ and $\partial u/\partial y$.

- (b) Using the Cauchy-Riemann equations, show that

$$\frac{\partial f}{\partial \bar{z}} = \frac{\partial u}{\partial \bar{z}} + i \frac{\partial v}{\partial \bar{z}}$$

is always zero. That is, a necessary condition for a function to be analytic is that it is “conjugate independent”.

Problem 2.3 (6 points)

We showed in class that for analytic functions the level sets of u and v always meet at right angles. In fact, something stronger is true.

Let $f(z)$ be analytic at a point z_0 and suppose $f'(z_0) \neq 0$.

- (a) Let $z(t) : [-1, 1] \rightarrow \mathbb{C}$ be a path in the complex plane such that $z(0) = z_0$ and it has a smooth derivative $z'(t)$. What is the tangent vector of this path at $t = 0$? What is the tangent vector of the path $Z(t) = f(z(t))$ at $t = 0$? Show that f rotates the tangent vector of v by a fixed angle.
 - (b) If I have two such paths $z(t)$ and $w(t)$ whose tangent vectors have a difference in angle of ϕ , what is the difference in angles of the paths $f(z(t))$ and $f(w(t))$?
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Problem 2.4 (4 points)

For the following harmonic functions, find a harmonic conjugate and corresponding analytic function.

- (a) $u(x, y) = 2x(1 - y)$
- (b) $u(x, y) = \sinh(x) \sin(y)$

Problem 2.5 (8 points)

- (a) Show that the function

$$g(z) = \sqrt{r}e^{i\theta/2}$$

is analytic on the domain $D = \{r > 0, -\pi < \theta < \pi\}$ and has derivative

$$g'(z) = \frac{1}{2g(z)}$$

- (b) Show that the composite function $G(z) = g(2z - 2 + i)$ is analytic on the half plane $\{\operatorname{Re}(z) > 1\}$ and has derivative

$$G'(z) = \frac{1}{g(2z - 2 + i)}$$

- (c) In part (a), can we make the domain D any larger and have g remain analytic? If so, give the largest domain of analyticity. If not, explain where and why analyticity fails. (Hint: what is the function g written in terms of z .)

End of homework